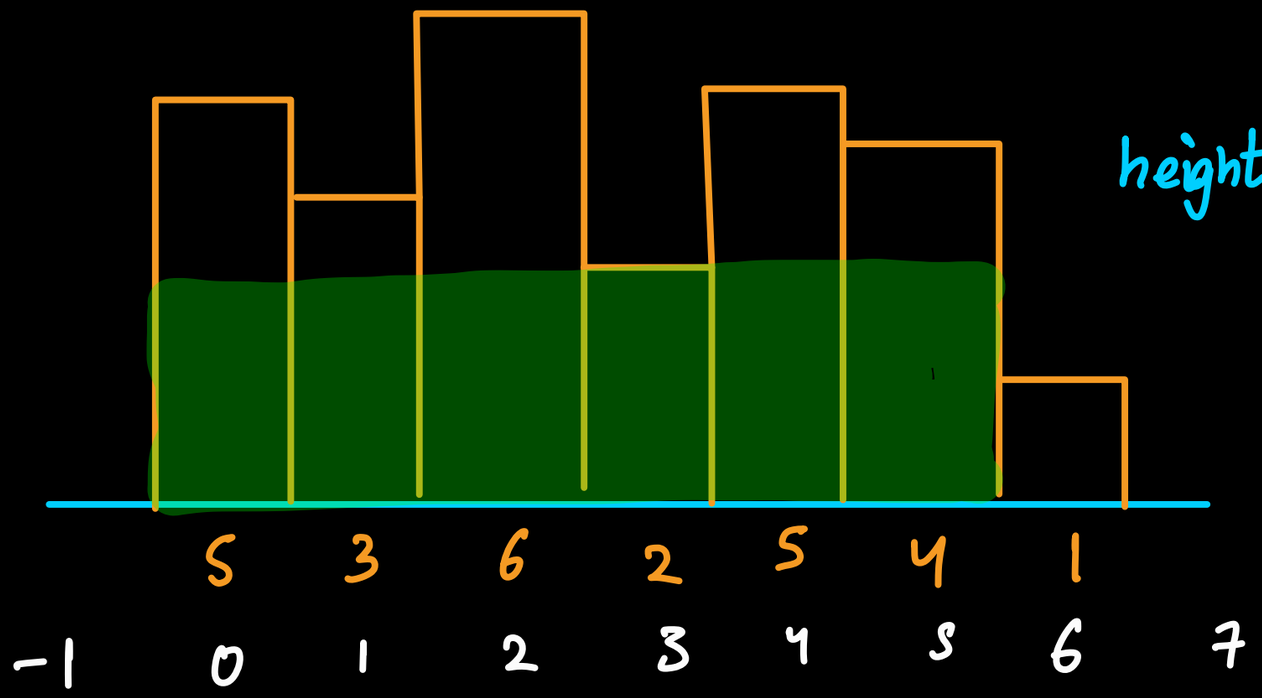
Ques: Largest Rectangle in histogram



heights[i]*(nse[i]-pse[i]-1)

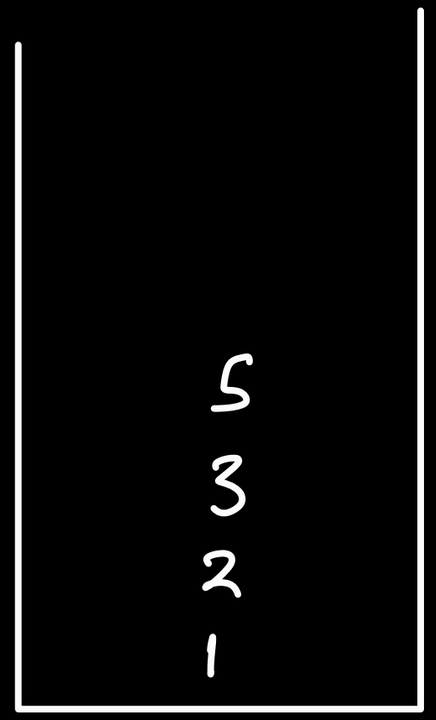
pse		-1	-1	1	-1	3	3	-1
nse		1	3	3	6	5	6	7
area		5	9	6	12	5	8	7

Ques: Largest Rectangle in histogram

next smaller element

arr =	5	3	6	2	5	4	1
nse =	3	2	2	1	4	1	-1

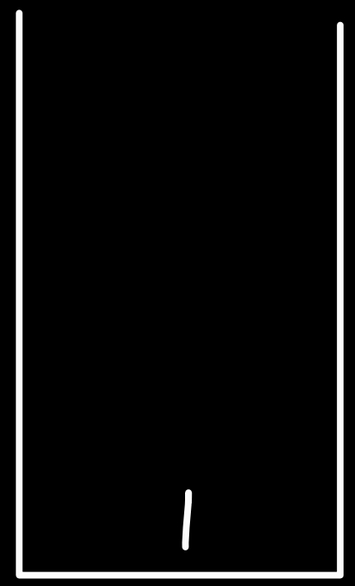
```
while (st.peek >= arr[i]) st.pop()
```



Ques: Largest Rectangle in histogram

Previous smaller element

arr =	5	3	6	2	5	4	1
pse =	-1	-1	3	-1	2	2	-1



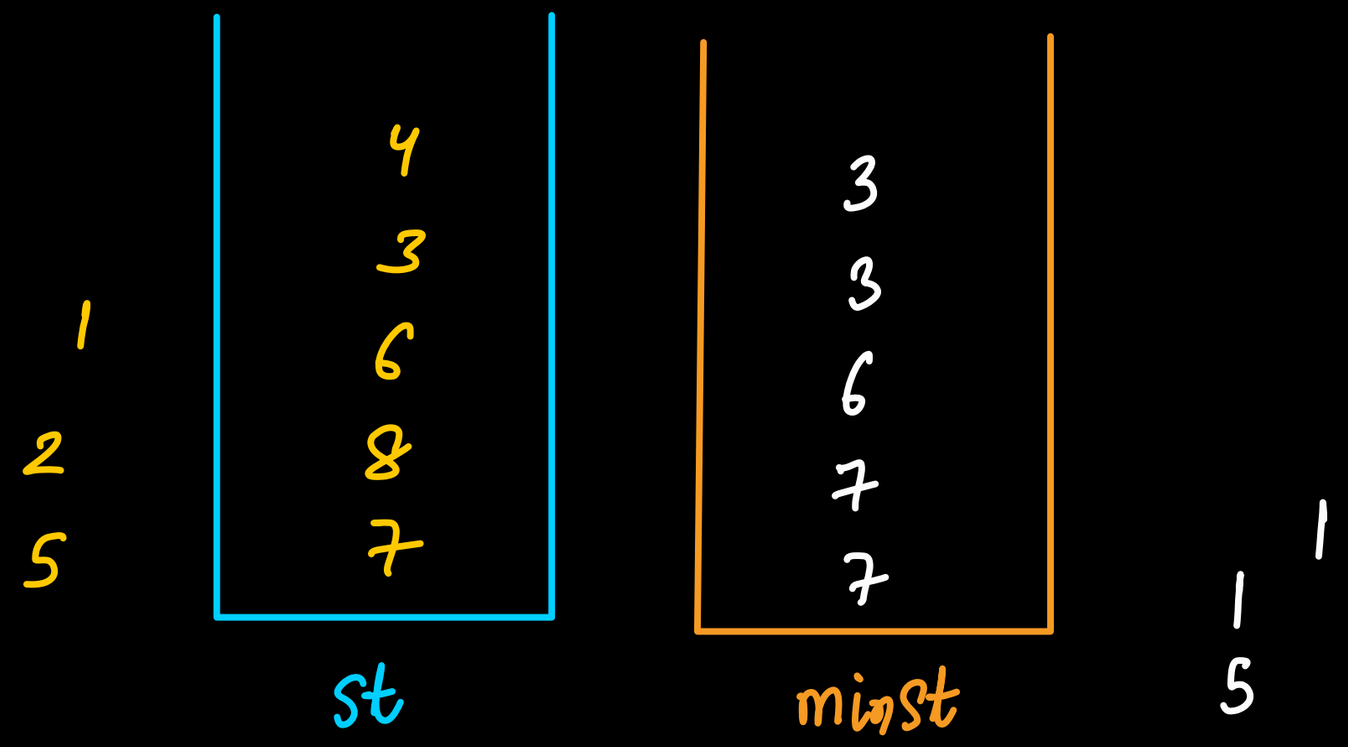
Ques: Maximal Rectangle

1	0	1	0	0
2	0	2	1	1
3	1	3	2	2
4	0	0	3	0

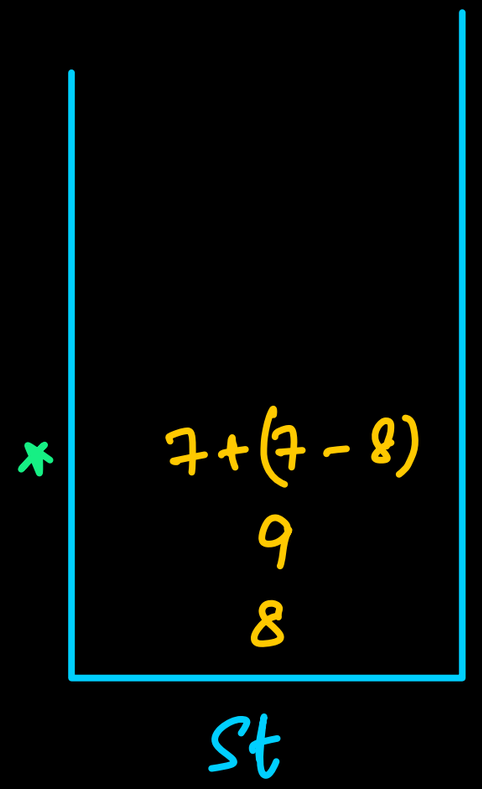
Steps : Make all col's prefix sum

Each row is an array. Send it to largest rectangle in Histogram method

Ques: Min Stack



Ques: Min Stack




```
int peek(){
  if(st.peek() < min)
  return 2*min - st.peek()
}
```

min = ~~8~~ 7

```
if (val < min){
st.push( 2*val - min)
  min = val
}
```

```
pop(){
  if(st.peek() < min)
  min = 2*min - st.peek
}
```